2014 Porcupine River Chum Salmon Telemetry Program

Prepared for:

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EDI Project No:

Project #14Y0354
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20 YEARS

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ENVIRONMENTAL DYNAMICS INC.

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EXECUTIVE SUMMARY

In 2013, the Department of Fisheries and Oceans transitioned the stock assessment of chum salmon in the Porcupine River from an enumeration weir on the Fishing Branch River to a sonar program located approximately 2 km downstream of Old Crow, YT. In 2011 and 2012 concurrent weir and sonar enumerations were completed so that future sonar based escapement estimates may be compared to historical weir based estimates. Radio tagging and telemetry studies on chum salmon were completed in 2013 to provide additional comparative data to allow for a better understanding of the relationship between weir and sonar counts. A similar radio tracking program was undertaken in 2014 to provide a 4th year of comparative data. As such, EDI Environmental Dynamics Inc. (EDI) captured and applied esophageal implant radio tags to 93 chum salmon during late August and September at the chum sonar site at Old Crow and conducted telemetry tracking flights throughout the Canadian portion of the Porcupine River watershed to relocate tagged chum on the spawning grounds.

In total, 73 of the 93 tags were accounted for through aerial tracking and captures in the local subsistence fishery. Six tags were recaptured and turned in by local fishers, two of which were turned in to the Vuntut Gwitch'in Government's (VGG) Natural Resources Department early enough in the season to be reapplied. Eight additional tags were located in the vicinity of Old Crow and were thought to have been captured in the local fishery and not turned in to VGG or died/regurgitated near the tagging site. These 8 tags were removed from the pool of tags that were successfully relocated; of the remaining 65 tags, 46.1% (30 of 65) were relocated in the Fishing Branch River upstream of the former weir location. The remaining 35 radio tagged chum salmon were found in the following locations: Fishing Branch River downstream of the former weir – 11, Porcupine River mainstem between David Lord Creek and the Whitestone River – 14, Old Crow River – 5, Bell River watershed including the upper Bell River and Rock River – 5 and the Bluefish River – 1.

The total 2014 Porcupine River chum salmon sonar count at Old Crow was 16,892 but 1,806 chum salmon were captured in the VGG subsistence fishery upstream of the tagging site. Chum salmon captured upstream of the tagging site were removed from the total sonar count, resulting in adjusted escapement of 15,086. Based on the percentage of tagged chum salmon found above a weir (46.1%), an estimated 6,955 chum salmon spawned upstream of the former weir during 2014. The data collected by this program provides additional insight into the spawning destinations of chum salmon in the Porcupine River, and will help in the development of appropriate escapement goals for the Porcupine River sonar program in the future.

ACKNOWLEDGEMENTS

Funding for this project was provided by the Yukon River Salmon Restoration and Enhancement Fund. Trix Tanner (Fisheries and Oceans Canada) provided technical support and reviewed the draft report. Mel Beshara (Fisheries and Oceans Canada) provided on-site support at the sonar and assisted with radio tag application. William Josie and Erica Tizya, of the Vuntut Gwitch'in Government Natural Resources Department provided general project direction and assistance with logistics, and collected tags recovered in the local fishery. Local VGFN member Wheeler Netro and Billy Bruce assisted with radio tag applications and provided the practical input to ensure the project's success.

AUTHORSHIP

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1 INTRODUCTION

1.1 PROJECT BACKGROUND AND GOALS

Salmon migrating up the Porcupine River through the Traditional Territory of the Vuntut Gwitch'in Government (VGG) are culturally important to VGG citizens. Old Crow, the only Canadian community on the Porcupine River, relies on the salmon fishery as a traditional food source and the subsistence harvest of salmon on the Porcupine River is an important component in the traditional lifestyle of Old Crow residents.

The Yukon River Panel was established to manage "transboundary" salmon stocks, which move through waters subject to both U.S. and Canadian fisheries management processes, in a cooperative forum. Three salmon species spawn in the Canadian portion of the Porcupine River: Chinook (*Oncorhynchus tshawytscha)*, chum (*Oncorhynchus keta)* and coho (*Oncorhynchus kisutch)*. Chinook and chum salmon are managed jointly through the Yukon River Panel's Joint Technical Committee (JTC) process; however, there is currently no joint management process for coho salmon.

Prior to 2013, a Fisheries and Oceans Canada (DFO) enumeration weir on the Fishing Branch River (a tributary to the Porcupine River) historically recorded annual passage rates between approximately 5,057 and 186,000 chum salmon, with an average annual return of approximately 44,000 (1998 to 2011; JTC 2012). This weir had been used to enumerate chum salmon in the Fishing Branch River, and to fulfill Canadian management obligations to provide an index of chum salmon escapement in the Canadian portion of the Porcupine River.

From 2003 to 2010, VGG conducted a combination of a mark-recapture and CPUE index program to provide an in-season estimate of chum passage on the Porcupine River passed Old Crow. In addition to providing passage estimates of chum, these programs spaghetti tagged numerous chum which provided information of the proportion of the run spawning upstream of the Fishing Branch River weir which was still operational at the time.

In 2009, the Vuntut Gwitch'in Government (VGG) initiated the Porcupine River sonar program, to allow for a more accurate in-season enumeration of chum salmon at a site located 2 km downstream of Old Crow. The Porcupine River sonar program has provided chum salmon escapement estimates annually since 2011. In response to the development of the sonar program, DFO discontinued the operation of the Fishing Branch River chum salmon enumeration weir after the 2012 chum salmon run. In 2013, VGG initiated the Porcupine River Chum Salmon Telemetry program to help enable the transition from weir to sonar based escapement goals as counts at the Porcupine River sonar site must be calibrated to counts from the Fishing Branch River weir. This was intended to quantify the proportion of chum salmon migrating past the sonar site which spawn upstream of the former weir site. The concurrent operation of the sonar and the Fishing Branch River weir in 2011 and 2012 provided two years of comparative data in this regard. In 2013, no weir count was available but radio tagging of chum salmon at the sonar site, and subsequent tracking of these salmon provided a means to estimate the proportion of fish that were counted on the sonar and were destined for spawning areas upstream of the weir. This proportion was used to estimate what the 2013 Fishing Branch River weir count would have been if the weir had remained in place.

Results from 2013 sonar and telemetry estimated that 74.1% of fish enumerated at the sonar site spawned upstream of the former weir site. This proportion is similar to the proportions documented in 2011 and 2012, when concurrent weir and sonar enumeration was completed. The proportion of fish passing the weir site was also within the range of historical mark-recapture/CPUE index programs in the watershed (completed from 2003 to 2010; EDI 2011).

In 2014, DFO took over the operation of the chum salmon sonar project near Old Crow while VGG continued the chum salmon telemetry program, to build on data from the 2013 telemetry program and to aid in the transition from weir to sonar based escapement goals. The 2014 chum telemetry project was coordinated by EDI on behalf of VGG and in partnership with DFO staff who were present for the duration of the chum run at the sonar site. DFO staff assisted with the project by overseeing the radio tagging of chum salmon captured when EDI technicians were not onsite.

The primary goal of the 2014 telemetry program, was to provide an estimate of the number of chum that passed the former weir site. The results from 2014 provide a fourth year of comparative tagging and/or concurrent sonar/weir data to help transition escapement estimates from the weir to sonar. A secondary goal of this project was to locate and document chum salmon spawning in other locations within the Porcupine River watershed, including portions of the Fishing Branch River downstream of the former weir. This program also collected information on chum salmon spawning habitat in these areas, and helped to develop local capacity to conduct fisheries work within the community of Old Crow.

1.2 RATIONALE FOR TRANSITION TO SONAR BASED ESCAPEMENT ESTIMATES

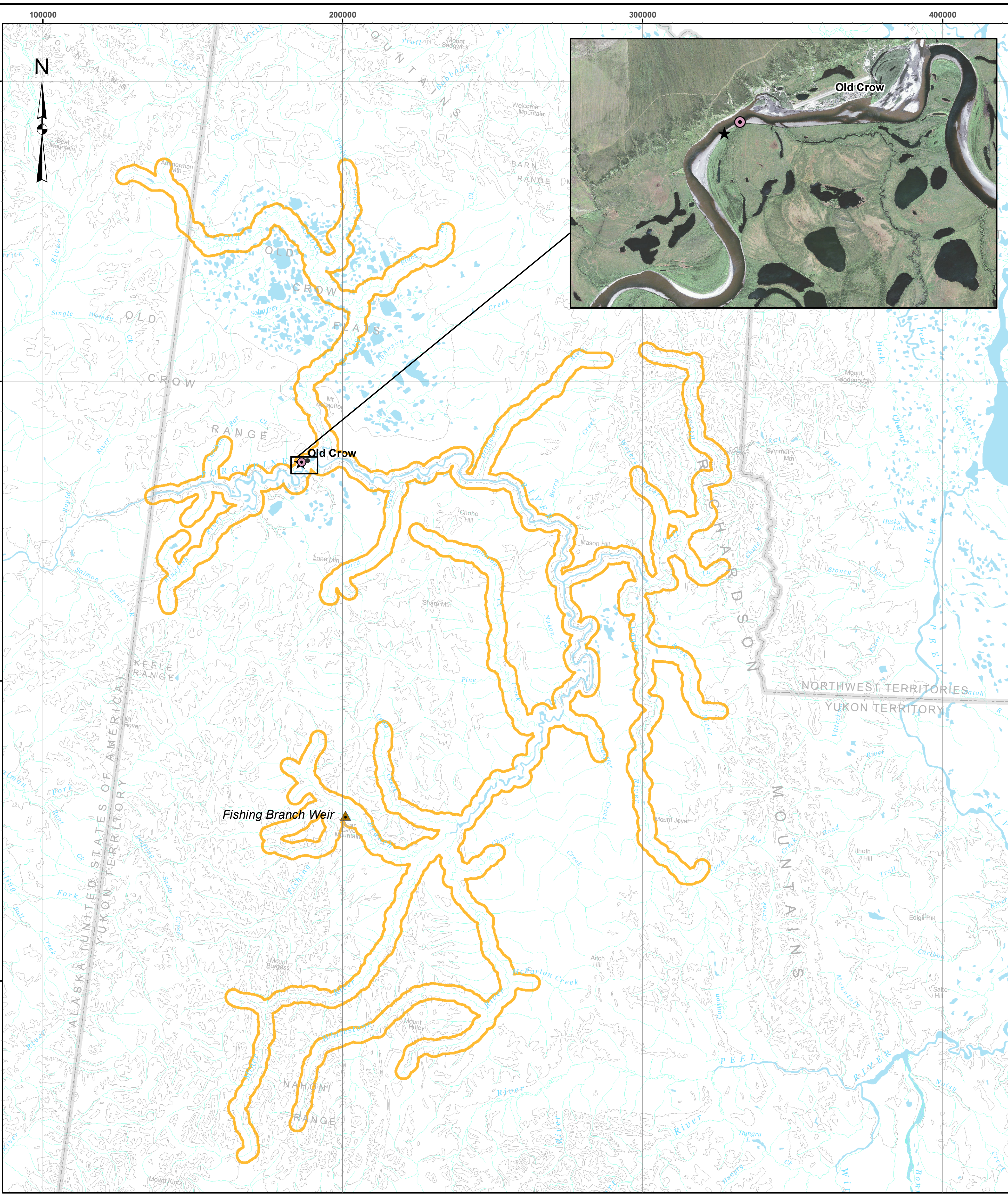
The primary motivation for the transition of the of the Porcupine River chum salmon stock assessment from weir based to sonar based escapement goals was related to the effective in-season management of this stock by fisheries managers. The VGG chum salmon subsistence fishery is the only harvest of the Porcupine River chum salmon in the Canadian portion of the river. The Fishing Branch weir was considered a terminal count as the weir site was a considerable distance upstream of the VGG subsistence fishery. The sonar site is located downstream of a large proportion of this fishery and therefore provides a more effective means to provide an in-season escapement of Porcupine River chum salmon. The data collected at the sonar site provides fisheries managers with a timely, in-season escapement data to help fisheries managers and resource users manage the local fishery in a manner that ensures a healthy Porcupine River chum salmon population in the future.

1.3 STUDY AREA

The Porcupine River is one of the largest tributaries in the Yukon River watershed, much of which is located within the Traditional Territory of the VGG. It has a number of large tributaries within Canada, including the Whitestone, Miner, and Fishing Branch rivers (Map 1). The only substantial Canadian settlement within the Porcupine River watershed is the community of Old Crow, located approximately 80 kilometres east of the Canada/U.S. border at the confluence of the Old Crow and Porcupine rivers. Old Crow has a population of approximately 300 people, mainly VGG citizens.

During 2014, all radio tags were applied in the vicinity of the Porcupine River sonar site; approximately 2 km downstream of the community of Old Crow (Map 1). No tags were applied upstream of Old Crow or downstream of the sonar site, where some tags were applied during the 2013 program.

Radio tag tracking flights were flown over much of the Canadian portion of the Porcupine River watershed (Map 1) and included all major tributaries including the Miner, Whitestone, Fishing Branch (including the North Fork), Bell and Old Crow rivers. Other medium sized tributaries surveyed including the Driftwood, Eagle, Rock and Bluefish rivers. A number of smaller tributaries were also surveyed including David Lord, Black Fox, Timber, Cody, Chance, McParlon, Pine and Caribou Bar creeks.



Legend

- Approximate Location of Tagging Site
- Approximate Location of Sonar Site
- Fishing Branch Weir - Inactive
- Extent of Telemetry Surveys (2014)

Overview map of the Porcupine River watershed showing chum radio tag application sites and extent of telemetry surveys

Data Sources
1,000,000 Topographic Spatial Data courtesy of Her Majesty the Queen in Right of Canada, Natural Resources Canada. All Rights Reserved.

Digital Elevation Model provided by Alaska State Geospatial Data Clearing House - http//www.asgdc.state.ak.us/.

1:1,000,000 National Topographic Database (NTDB) provided by Geomatics Yukon - Yukon Government via online source (Corporate Spatial Warehouse) www.geomaticsyukon.ca.

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Disclaimer
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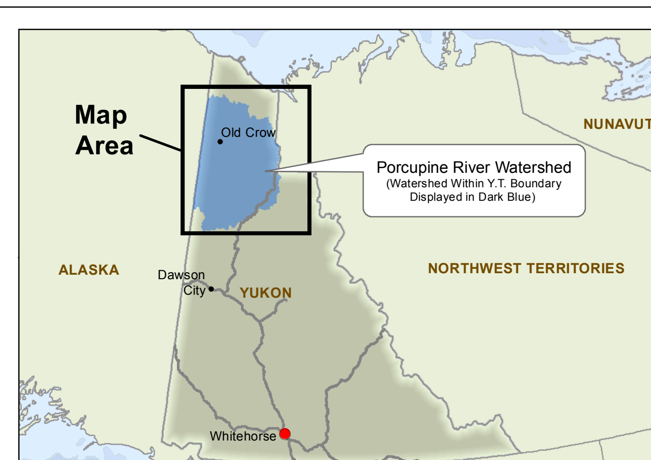
0 5 10 20 30 40 50
Kilometres

Map Scale 1:1,300,000 (printed on 11 x 17)
Map Projection: North American Datum 1983 CSRS Yukon Albers

Drawn: LG	Checked: BSn/MP	Date: 16/02/2015	MAP 1



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2 METHODS

2.1 CHUM SALMON RADIO TAGGING

The Porcupine Sonar program was operational from August 12 to October 1, 2014 and radio tagging of chum salmon occurred in the vicinity of the sonar site from August 25 to September 29, 2014. EDI applied tags from August 25-28 (14 tags), September 9-11 (11 tags) and September 15-21 (39 tags) assisted by local VGG technicians working on the chum salmon sonar program. The remaining 31 tags were applied by the VGG technicians at the sonar program under the supervision of DFO staff working at the sonar. Tags were planned to be proportionally to male and female fish; however, due to low fish capture rates, this was only done when possible.

Chum salmon were captured via set netting and beach walk (near shore) drift netting. Beach walks were deployed just downstream of the Porcupine River sonar site and set nets were conducted at various sites within 1-2 km of the sonar site (Map 1). Many of these sites had been used during previous chum mark-recapture programs (Snow 2010) and were known to be effective fishing sites at various water levels; it is also important to note that there are no known chum salmon spawning areas in the vicinity of the sonar site. Set nets were constantly monitored and checked regularly (approximately hourly) or when fish were visibly captured in the net. The gillnets used were 100 feet in length by approximately 10 feet deep (mesh depth of 29, using cable salmon net mesh); nets were single panels, with mesh sizes of 4.5, 5 and 6 inches.

Captured chum salmon were quickly removed from the net and placed in a water filled tote to recover. Fish were observed for any signs of injury (e.g. lacerations, bleeding from gills), and healthy fish were selected for tagging. Scale samples were collected from all captured chum and delivered to DFO for processing; any injured chum salmon were sexed, measured, and released untagged. All other captured fish were identified to species, measured and released.

Chum salmon selected for tagging were fitted with ATS (Advanced Telemetry Systems) model F1840B esophageal implant tags. Esophageal implant tags have two advantages over the dorsal implant tags used in 2013. The first advantage is that esophageal implant tags are internal and following the 2013 project, it was thought that dorsal implant tags may increase the likelihood of capture in the local subsistence gill net fishery as the external tag is easily entangled in the nets. The second advantage is that if an esophageal implant tag is recovered from a harvested fish it can be reapplied while dorsal implant tags are one-time use only and cannot be reapplied. In addition to the radio tag, a secondary numbered orange "spaghetti" tag was applied to each radio tagged chum (Photo 1). Spaghetti tags were meant to make radio tagged fish more visible to local fishers; tags were brightly colored and fishers were accustomed to looking for tags due to previous chum salmon research programs on the Porcupine River. Following each successful tag placement, fish were allowed to recover once more and were inspected for any signs of injury or unusual stress prior to release.

Photo 1. Local VGG technician with a female chum salmon tagged with an esophageal implant tag and posterior secondary spaghetti tag; August 26, 2014.

A total of 93 radio tags on four different frequencies were utilized for this project, as shown in Table 1; two of the ninety three were also reapplied following recapture in the local VGG subsistence fishery. One additional was retained as a tester tag to allow for the testing of the radio receiver and antennas prior to tracking flights. Each tag was coded with a unique identifier; a full list of tag frequency and unique tag identifiers is provided in Appendix A. Tags were also encoded with a mortality sensor, to help determine whether tracked chum had reached their terminal spawning destinations. Mortality sensors were pre-set to activate if no movement was registered by the tag for 24 hours or more.

Table 1. Frequencies of radio tags applied to chum salmon at the Porcupine River sonar site in 2014.

Tag Frequency	Number of Tags
164.187	25
164.206	24
164.236	25
164.506	19

2.2 TRACKING RADIO TAGGED CHUM SALMON

Aerial tracking of radio tagged chum was conducted over a 4 day period (October 23 to 26, 2014) by an EDI technician; the areas flown on each day of the survey are shown in Table 2. The Fishing Branch River mainstem was flown on the first and last day of surveys, to ensure that all tagged fish that spawned upstream of the weir were detected.

Table 2. Summary of chum salmon telemetry surveys in the Porcupine River watershed in 2014.

Date of Tracking Flight	Survey Locations
October 23, 2014	• Miner River, including Fishing Creek • Whitestone River, including north fork • South fork of the Fishing Branch River • North fork of the Fishing Branch River
October 24, 2014	• East fork of the Whitestone River, including Chance and McParlon Creeks • Crow River, including Timber and Blackfox Creeks • Porcupine River mainstem from the Whitestone confluence to Old Crow
October 25, 2014	• Porcupine River mainstem from Old Crow to Canada/U.S. border • Caribou Bar Creek • Bluefish River • David Lord Creek • Driftwood River • Bell River, including the Waters and Little Bell River
October 26, 2014	• Eagle River • Rock River • Pine Creek • South fork of the Fishing Branch River • North fork of the Fishing Branch River

Aerial tracking was conducted using a Cessna 172 chartered from Whitehorse. Radio antennas were mounted to each wing strut of the plane, to allow for optimal detection of tags on either side of the plane's flight path. Tracking was completed using an ATS model R4520C GPS equipped receiver; the same receiver unit was used on all flights. The frequencies of applied tags were input into the receiver, and the receiver was set to cycle through each frequency at a scan rate of 3 seconds per channel. Prior to the tracking flights, the accuracy of the receiver's GPS was tested against a handheld GPS unit of known accuracy and found to be within ±10 m.

While tracking, the plane was flown as low as possible (300 to 500 feet), at speeds ranging from 100 to 130 km/h (the slowest possible speed, dependent on wind conditions). When a tag tone was heard by the technician, the frequency was locked using the receiver's "hold" function and the tag was decoded. When required, the pilot was asked to fly a loop to allow the receiver sufficient time to detect the tag. On October 23 and 26, the Fishing Branch River (where many tagged chum were expected to have spawned) was flown four times, with one frequency on 'hold' for each pass. This facilitated decoding of all tags in the area, and helped ensure that no tags were missed while the receiver was cycling through frequencies.

A total of six tags were recaptured in the local fishery and turned in to VGG's Natural Resource department. Two tags were recovered early in the run and were redeployed to maximize the sample size of tagged chum.

2.3 DETERMINING LOCATIONS AND MORTALITY OF RADIO TAGGED CHUM SALMON

The locations of all chum salmon detected during radio tracking flights were determined from the recorded telemetry data. For each tag that was decoded, the R4520C radio receiver recorded the date, time, frequency, tag identifier, signal strength and GPS coordinates (UTMs). After each survey, this data was downloaded from the receiver and exported to excel. Data from all days of the survey was compiled together for further analysis.

Each tag was set to continually broadcast its data, and therefore several data records were collected for most tags. To determine the final position of each tagged chum salmon, the data records from all surveys were sorted by frequency, tag identifier and signal strength. For each tag, the data record with the highest signal strength was taken as the best approximation of the tag's location.

To determine the status of a tag's mortality sensor, the last time stamp record was examined and the mortality sensor data was analyzed for each tag individually. As the mortality sensor can switch "on" and "off" over time, an average of all the latest recorded mortality codes were taken for each fish. If a fish was flown over multiple times, only the latest flight's records were analyzed. Taking the mean of the mortality codes enabled comparison between records that had variable mortality readings. The three possible codes are zero, indicating very much alive, two indicating inactive and likely dead, and one indicating an intermediate activity level and interpreted as not clearly alive or dead. When averages were taken, every individual with a mean mort code of less than one was assumed to be "alive", while averages greater than one are assumed "dead". Individuals with a mean mort code of one were categorized as "unclear".

In most cases, the mortality sensors provide a good indication of whether a tagged fish is alive or dead; however, false positive are possible. For example, a chum carcass that is caught on larger woody debris in an area of high current could be moving enough to maintain the mortality sensor in the "alive" position, erroneously indicating this fish was still alive. For this purposes of this project, the status of the mortality sensors on tagged chum salmon were taken as indicator of the vitality of each tagged chum, but were not taken as proof that a particular chum salmon was alive or dead.

3 RESULTS

3.1 CHUM SALMON TAGGING

3.1.1 Fish Capture Data

A summary of combined weekly drift and set net catches conducted by EDI and DFO over the duration of the sonar program are presented in Table 3 and Appendix B. A total of 203 fish were captured including 196 chum salmon and 7 freshwater fish (1 Arctic grayling, 1 inconnu and 5 whitefish sp.).

Table 3. Summary of 2014 weekly drift and set netting fish captures at the Porcupine River sonar site.

Program Week	Chum Salmon Captured	Other Fish Species Captured	Total Fish Captured	Chum Proportion (%)
August 25 - 31 [A]	19 [A, B]	0	19	100
September 1 - 7 [A]	4 [A]	NA	NA	NA
September 8 - 14	15	1	16	94
September 15 - 21	39	5	44	89
September 22 - 28	52	1	53	98
September 29 - October 1	67	0	67	100
Total	**199 [A]**	**7**	**203**	**96.6**

[A] Fish sampling data from August 27 to September 7 was lost in the field; however, the number of chum tagged during each week was retained.

[B] The number of chum captured from August 25 to 31 includes only the individuals tagged (none released untagged during this period).

3.1.2 Radio Tag Deployment Data

A summary of weekly radio tag deployment totals are shown in Table 4. Tag application rates were spaced out through the run. Detailed drift netting and set netting data, and information on the location and date that each radio tag was applied can be found in Appendix B. Note that the two recaptured tags which were re-applied are only included once in the following table (initial deployment removed).

Table 4. Summary of weekly chum salmon radio tag deployments on the Porcupine River in 2014.

Program Week	Number of Radio Tagged chum	Proportion of Total Tags (%) [A]
August 25 - August 31	19	20.4
September 1 - September 7	4	4.3
September 8 - September 14	10	10.8
September 15 - September 21	38	40.9
September 22 - September 28	20	21.5
September 29	2	2.2
Total	**93**	**100.0**

The sex ratios and fork lengths of radio tagged chum salmon that were captured between August 24 and October 4, 2014 are shown in Table 5. The average fork length of captured male chum salmon was 68.8 cm, while the average fork length of female chum salmon was 62.1 cm (Table 5). Male chum salmon were tagged more frequently than females due to a higher capture rate and accounted for 79.3% of all chum tagged. The lengths and sex ratios are similar to the results of past chum salmon netting programs on the Porcupine River (EDI 2011, Snow 2010). At the time that this report was written, analysis of scale samples taken from captured chum salmon was not yet completed by DFO.

Table 5. Weekly summary of sex and fork length data from chum salmon radio tagged in the Porcupine River in 2014.

Program Week	Male			Female		
	Total Weekly Tagged	% of Weekly Total	Mean Length (cm)	Total Weekly Tagged	% of Weekly Total	Mean Length (cm)
August 25 - August 31 [A]	12	86	NA	2	14	NA
September 1 - September 7 [A]	NA	NA	NA	NA	NA	NA
September 8 - September 14	10	91	64.5	1	9	62.0
September 15 - September 21	15	28	71.1	24	72	62.6
September 22 - September 28	11	55	67.3	9	45	62.3
September 29 - October 3	0	0	NA	2	100	62.0
Total tagged with age/length data	**48** [B]	**-**	**-**	**38**	**-**	**-**
Program Mean	**-**	**56**	**68.0**		**44**	**62.5**

[A] Fish sampling data from August 27 to September 7 lost in the field.

[B] Seven addition chum of unknown sex were tagged without fork lengths recorded.

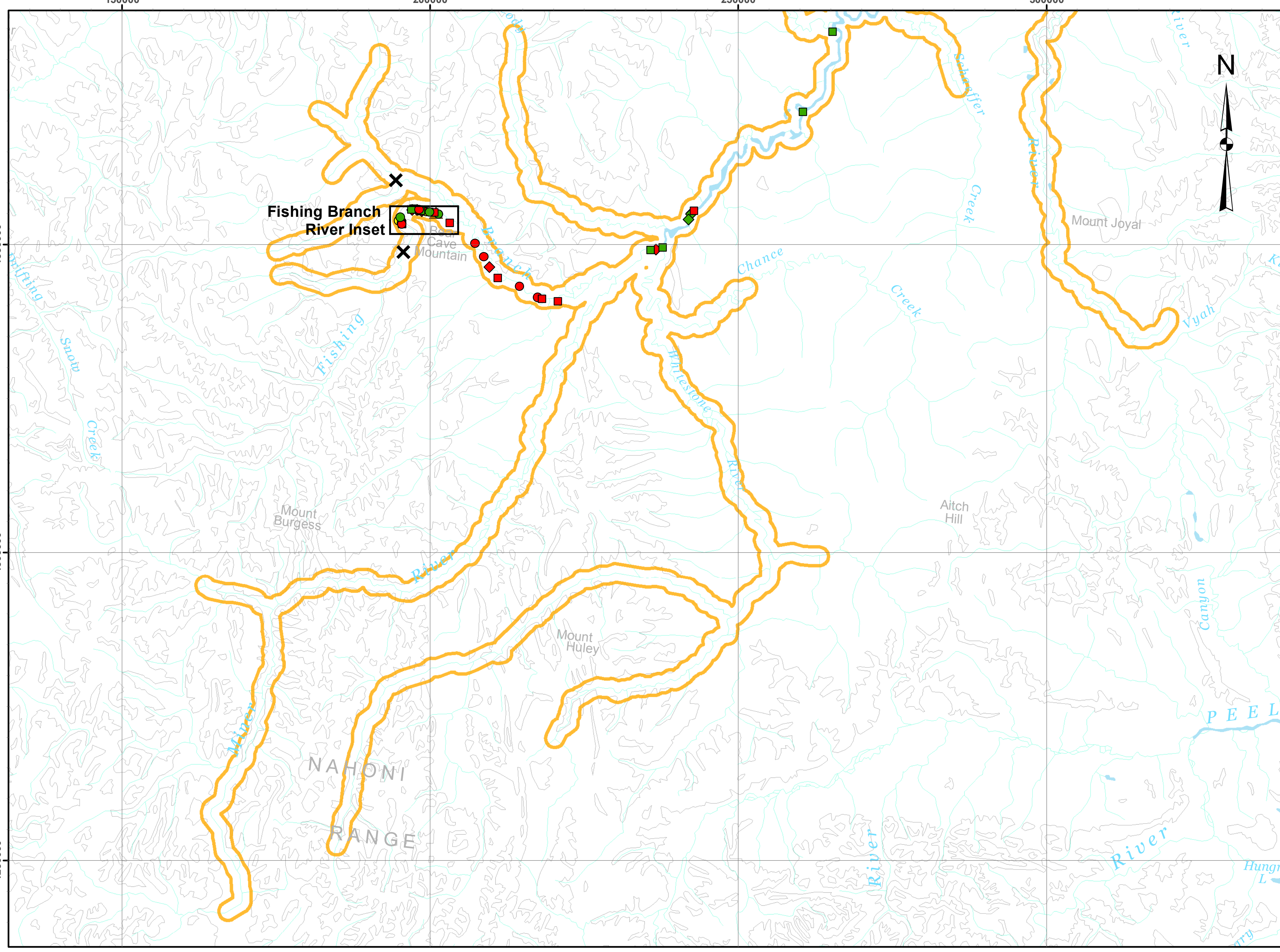
3.2 RELOCATIONS OF RADIO TAGGED CHUM SALMON

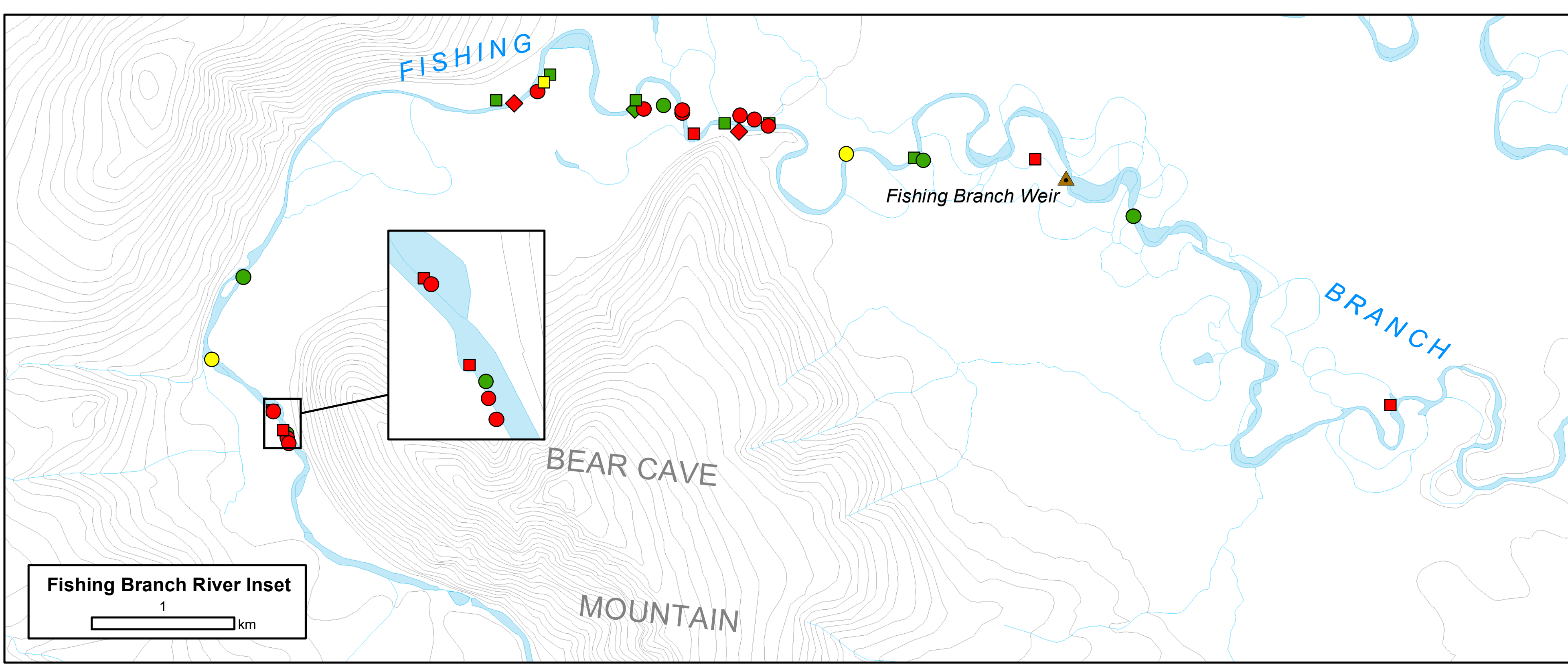
Chum salmon radio tracking flights conducted in October 2014 located 73 tags that were applied during the 2014 tagging program. When including the additional 4 tags recovered by local fishers 77 of the 93 applied tags were accounted for. The locations of relocated tags are summarized in Table 6; specific coordinates for each tag can be found in Appendix C. Surveys of the Fishing Branch River located 30 tagged chum salmon upstream of the former weir site (Table 6; Map 2). Eleven tags were located in the Fishing Branch River downstream of the weir site and no tags were located in the North Fork of the Fishing Branch River.

A considerable number of tags (15) were located in the Porcupine River mainstem, with the remaining tag detections in the vicinity of Old Crow, as well as the Bell, Bluefish and Crow rivers (Table 6). As there are no known suitable chum salmon spawning areas in the vicinity of Old Crow, tags detected in this area were likely caught in the local subsistence fishery and not turned in, tagging related mortalities, or tag regurgitations.

Table 6. Summary of locations of chum salmon radio tagged in the Porcupine River in 2014.

Radio Tag Location	Number of Tags Detected
South Fork of the Fishing Branch River, upstream of the former weir site	30
South Fork of the Fishing Branch River, downstream of the former weir site	11
Porcupine River mainstem, from Whitestone River confluence to Schaeffer Creek	8
Porcupine River mainstem, from Schaeffer Creek to mouth of Bell River	5
Porcupine River mainstem, between mouth of Driftwood River and David Lord Creek	1
Old Crow, and near vicinity (not including recaptured tags turned in)	8
Crow River	5
Bell River, including Rock River and Little Bell River	4
Bluefish River	1
TOTAL FOUND	**73**





Legend

- Fishing Branch Weir - Inactive
- End of Wetted Channel
- Extent of Telemetry Surveys (2014)

Radio Tag Location

Sex Code		Mortality Code	
◻	Female	Green	Alive
○	Male	Red	Dead
◇	Unknown	Yellow	Unclear

Locations of radio tagged chum salmon in the southern portion of the Porcupine River watershed in October 2014

Data Sources
1,000,000 Topographic Spatial Data courtesy of Her Majesty the Queen in Right of Canada, Natural Resources Canada. All Rights Reserved.

Digital Elevation Model provided by Alaska State Geospatial Data Clearing House - http//www.asgdc.state.ak.us/.

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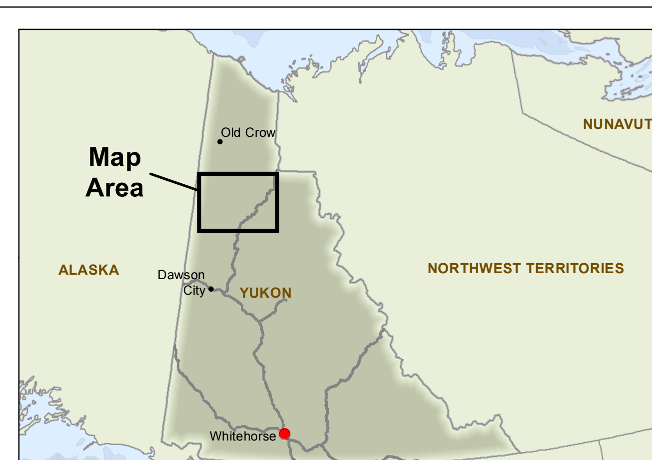
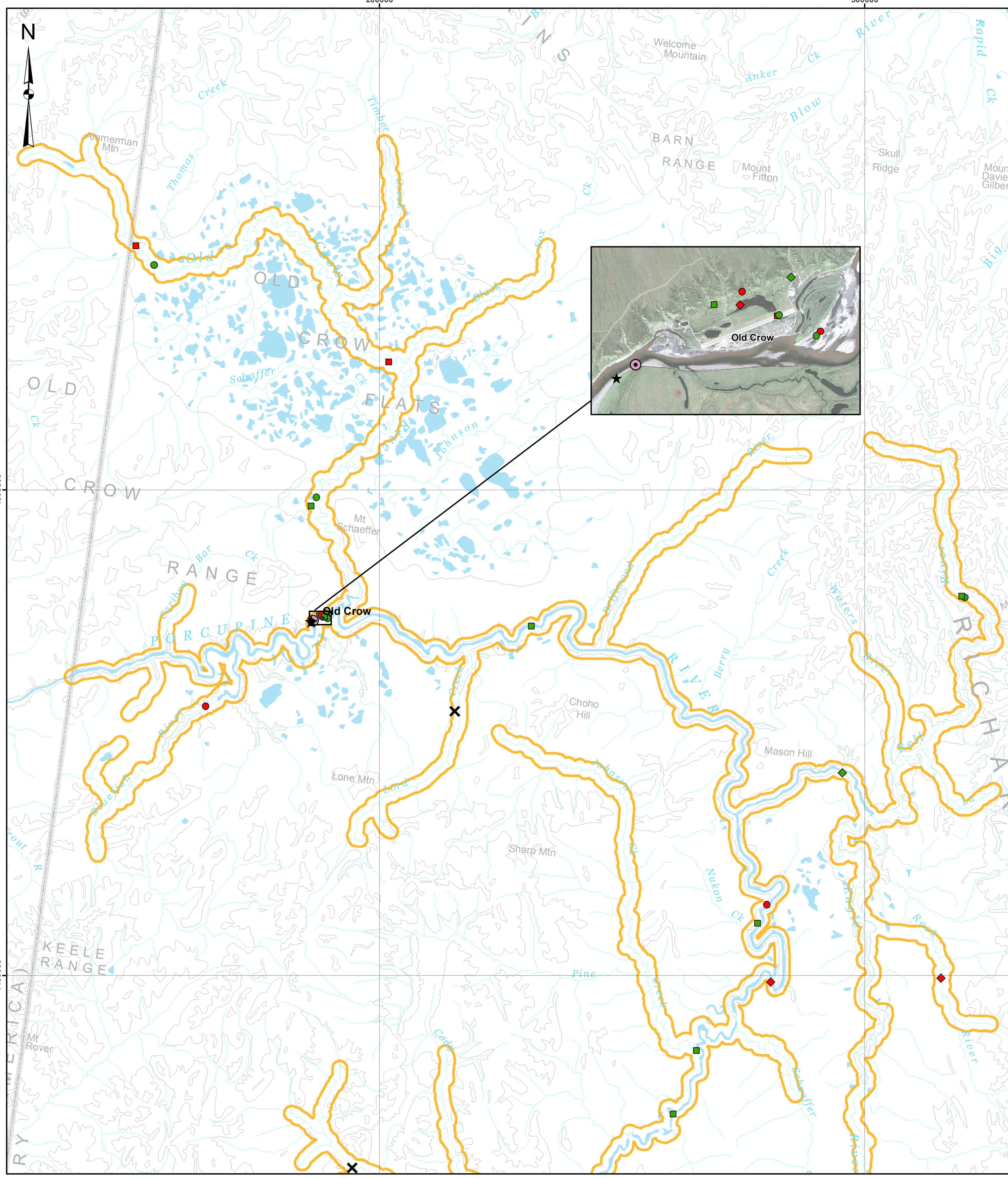
0 5 10 20 30 40 50
Kilometres

Map Scale 1:800,000 (printed on 11 x 17)
Map Projection: North American Datum 1983 CSRS Yukon Albers

Drawn:	Checked:		
LG	BSn/MP	Date: 06/02/2015	**MAP 2**





Legend

- End of Wetted Channel
- Approximate Location of Sonar Site
- Approximate Location of Tagging Site
- Extent of Telemetry Surveys (2014)

Radio Tag Location

Sex Code		Mortality Code	
□	Female	Green	Alive
○	Male	Red	Dead
◇	Unknown	Yellow	Unclear

Locations of radio tagged chum salmon in the northern portion of the Porcupine River watershed in October 2014

Data Sources
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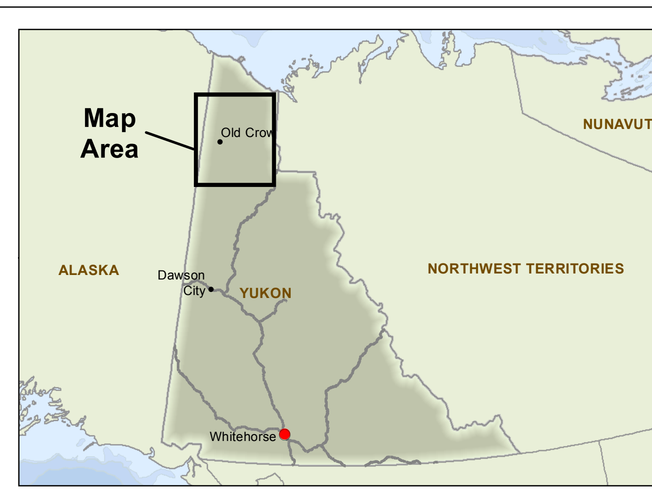
Drawn: LG	Checked: BSn/MP	Date: 16/02/2015	MAP 3



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3.2.1 Mortality Sensor Status

The status of the mortality sensors for the detected tags (excluding tags in the vicinity of Old Crow) is summarized in Table 7. More than half of the detected tags were found with their mortality sensors 'on' (Table 7), meaning that no movement had occurred in over 24 hours. In the lower part of the Fishing Branch River most of the tags had "dead" mortality sensor status. In contrast, the mortality sensors in the tags found in the Porcupine River mainstem and the Bell and Crow rivers were mostly "alive". Given the period of time between tagging and the telemetry flights (a minimum of 24 days), these fish should have been located on the spawning grounds. The most likely explanation for the tags giving off the "alive" signal is carcass and/or tag drift in the river.

Table 7. Summary of mortality sensor status for radio tagged chum salmon in the Porcupine River in 2014 (excludes tagged chum salmon recaptured in local subsistence fishery).

Location	Number of Tags with Mortality Status Dead	Number of Tags with Mortality Status Alive	Number of Tags with Mortality Status Unclear	Percentage of 'Dead' Tags (%)
Fishing Branch River – Upstream of weir site	17	12	3	53
Fishing Branch River – Downstream of weir site	10	1	0	91
Porcupine River Mainstem	5	9	0	36
Crow River	2	3	0	40
Bell River	1	3	0	25
Bluefish River	1	0	0	100
TOTAL	**36**	**28**	**3**	**54**

3.3 Spawning Habitat Charateristics of Surveyed Watersheds

The suitability of spawning habitat for chum salmon in the surveyed tributaries of the Porcupine River was described extensively in the 2013 Porcupine River Chum Salmon Telemetry report (EDI 2014b). Refer to EDI 2014b (Section 4) for detailed habitat description of the following tributaries: Fishing Branch River North Fork, Miner River, Whitestone River, Porcupine River Mainstem, Bell River, Driftwood River, David Lord Creek, Bluefish River and Crow River. The areas of the Fishing Branch River that are upstream of the former weir are known to be productive chum salmon spawning habitats, and are well documented. These habitats are not discussed further in this report. The descriptions below focus on streams where tags were detected for the first time during 2014 including: the upper Bell River, the Crow River mainstem and the Rock River. It is also important to note that the tags relocated in the lower Bell River, lower Crow River and the Porcupine River mainstem between Old Crow and Schaeffer Creek may represent mortalities or tag regurgitations. Although spawning in these areas cannot be ruled out, the habitat in these areas does not appear to be suitable for spawning (i.e., no evidence of upwelling).

Chum salmon are known to spawn in areas of upwelling water which are typically warmer than surrounding areas (McPhail 2007) when choosing spawning locations. This preference indicates that groundwater discharge areas are important in the selection of spawning habitat by chum salmon. Existing information indicates that Porcupine River chum spawn in areas with notable groundwater discharge including the

Fishing Branch River and other areas such as the Bluefish River and the mouth of David Lord Creek. Coho salmon spawning in the watershed has also been shown to be associated with groundwater discharge areas (Schonewille and Anderton 2006; EDI 2008). Conversely, Chinook salmon spawning in the Porcupine River appears to be less dependent on upwelling water, as Chinook salmon have been observed spawning in many areas (e.g. the Miner River; EDI 2014a) where no evidence of groundwater was found during winter telemetry flights (Schonewille and Anderton 2006; EDI 2008).

In the winter, the presence of groundwater upwelling areas is often indicated by ice free river/stream channels. In many locations that were surveyed in October 2014, ice free areas of stream channel were observed in the vicinity of locations where tags were located. These observations are discussed in the relevant sections below for area. Past telemetry data from radio tagged coho salmon collected during the winter months is also discussed in the sections below, where relevant. Chinook salmon data is not discussed, as the differing life history of this species suggest that they select different spawning habitat than coho and chum salmon in the Porcupine River.

3.3.1 Upper Bell River

The mainstem Bell River was surveyed in its entirety from its source in the Richardson Mountains to the confluence with the Porcupine River, a distance of approximately 185 km. During the time of the survey, the majority of the river was ice covered with the exception of the upper portion of the watershed (i.e., upstream of the Little Bell River confluence). It was in this area that two radio tagged chum were relocated during the October 25, 2014 aerial telemetry flight (Photo 2). These tags were located approximately 12 km upstream of the Little Bell River confluence and were in an area of extensive open water, indicating that upwelling groundwater was likely present.

The upper Bell River has long be considered a probable spawning location for salmon in the Porcupine River watershed; however, the two radio tagged chum found during 2014 constituted the first confirmed presence of any species of salmon in this area. This information provides the first confirmed presence of any species of salmon in the Bell River watershed as pre-existing information was limited to historical catch records of chum salmon at Lapierre House on the main stem Bell River. However, the spawning destination of these chum salmon was previously unknown (Anderton and Frost 2002).

Ground based investigations conducted by the North Yukon Renewable Resources Council and VGG during 2003 (CRE-15-03; Anderton 2003) in the upper Bell River did not capture any salmon in this area; however, the presence of suitable spawning gravels was noted in the same location where the 2014 tags were located. The water temperatures during the July 2003 investigations were also very cool (5 to 6°C) despite warm mid-summer air temperatures and are also indicative of a groundwater discharge area. Note that the one radio tag relocated in the lower Bell River does not likely represent a spawning location as previous investigations (EDI 2014b) have determined that this area is not suitable for salmon spawning.

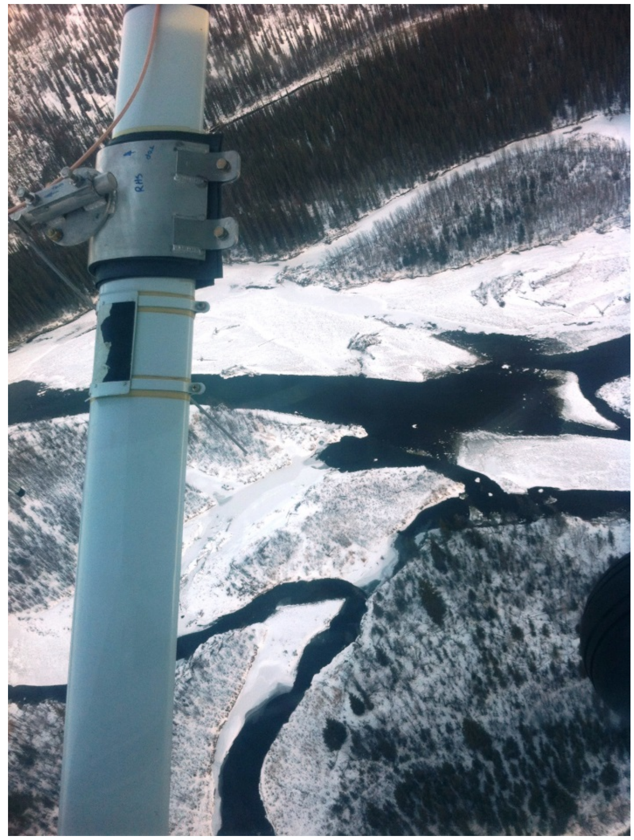

Photo 2. Aerial view of the upper Bell River where two radio tagged chum salmon was relocated on October 25, 2014. Note the extensive open water suggesting that this is a groundwater discharge area.

3.3.1 Rock River

The Rock River is a medium sized tributary which originates in the Richardson Mountains to the east of Eagle Plains (Dempster Highway) and flows into the Bell River approximately 40 km upstream from the Porcupine River. During the October 26, 2104 aerial survey, the Rock River was surveyed from the confluence with the Bell River upstream to the confluence of White Creek, near the Dempster Highway crossing.

One radio tagged chum salmon was relocated on October 26, 2014 in the middle reaches of Rock River, approximately 10 km downstream of the White Creek confluence. This information provided the first documented presence of any species of salmon in the Rock River which was not previously suspected to provide suitable salmon spawning habitat. Ground based investigations conducted by the North Yukon Renewable Resources Council and VGG in the lower Rock River during 2003 (CRE-15-03; Anderton 2003) did not note any suitable salmon spawning habitat in the lower Rock Rover. However, the portion of the river where the radio tag was relocated was not surveyed during this program. The portion of the Rock River near the White Creek confluence (i.e., near where the radio tag was relocated) may provide suitable spawning habitat and this may represent a potential spawning area. The presence of open water in this area (Photo 3) did not appear to suggest the presence of groundwater discharge but rather a velocity induced open water lead. The relocation of a single tag in this area may also represent a stray chum and flying this same area during future telemetry project may provide additional information on the likelihood of spawning in this portion of the Rock River.

Photo 3. Aerial view of the Rock River near the White Creek confluence where a radio tagged chum salmon was relocated on October 26, 2014.

3.3.2 Old Crow River Mainstem

The Old Crow River mainstem was surveyed in its entirety from its source near Ammerman Mountain on the Yukon/Alaska border to its confluence with the Porcupine River at the community of Old Crow. During the October 24, 2014 aerial survey, the majority of the river was ice covered with the exception of the upper reaches which contained extensive open water and a clean gravel bottom. These observations are consistent with those from the 2007 coho telemetry project which noted extensive open water in the vicinity of Ammerman Mountain during mid-winter (EDI 2008).

During the 2014 telemetry flights, a total of 5 radio tags were relocated within the Old Crow River mainstem. Two tags were located in the upper reaches of the river near the Yukon/Alaska border and were directly downstream of the area of extensive open water (probable groundwater discharge area). Considering the apparent quality of habitat in this area, the location where these tags were located represents a probable chum spawning area. Also of note, historical field investigations in this area (Steigenberger 1975a) estimated in excess of 10,000 freshwater fish overwintering in this vicinity. Traditional knowledge and other local reports (Anderson and Frost 2002, 2003) have noted the presence of chum salmon in the Old Crow River; however, the relocation of these two radio tags in a probable spawning location provides the first strong evidence of spawning in the watershed.

The remaining three tags in the mainstem were located in the vicinity of Mt. Schaeffer in the lower reaches of the river (approximately 35 km upstream of Old Crow; 2 tags) and near the confluence of Black Fox Creek (1 tag). During the 2007 coho telemetry project, no obvious signs of groundwater were noted in these areas and historical field investigations by Steigenberger (1975b) noted that substantial portions of the middle and lower portions of the Old Crow River were found to be frozen solid to the riverbed. Although spawning in the middle and lower reaches of the Old Crow cannot be ruled out, is appears unlikely that spawning occurs in this area.

4 DISCUSSION

4.1 DEVELOPMENT OF LOCAL CAPACITY

An important goal of the 2014 chum radio tagging program was to continue to develop local capacity to conduct fisheries work within the community of Old Crow. While a number of past programs (e.g., mark-recapture, CPUE index and sonar programs) have helped to develop fisheries technical capacity within the community, this particular program provides several important advantages:

- The program offered approximately 6 weeks of fisheries related work to two individuals when combined with the concurrent chum sonar program;
- The skills needed to apply the radio tags and conduct drift netting provided new learning opportunities for local technicians; and,
- The program provides a means to employ skilled fisheries technicians in the community, while also providing experience to younger community members who have not been involved in past programs.

Two local field technicians were trained and participated in this program during 2014. Field technicians conducted a portion of the tagging work independently from the biologists, and were essential to safe and successful drift netting. The knowledge of the theory and rationale of radio tagging fish is a specialized skill and there is a considerable amount of interest within the community of Old Crow to pursue further stock assessment programs, especially for the Porcupine River Chinook salmon. If the additional radio tagging studies are conducted in the Porcupine River watershed, there is an existing technical understanding of radio tag application tecnhiques.

4.2 ADJUSTED TOTAL RADIO TAG NUMBERS

The results of the chum salmon radio tag tracking confirm that the Fishing Branch River was the major spawning destination for chum salmon in 2014. Of the 93 tags that were applied during the 2013 field program, 30 tags were found in the Fishing Branch River in spawning areas upstream of the former DFO enumeration weir. Of the 11 tags relocated downstream of the former weir site, 10 of the tags were broadcasting a mortality signal; thus indicating that these tags had not moved for 24 hours. This finding supports that the fact that the timing of the 2014 aerial telemetry flight was sufficient to ensure that all tags were located on the spawning grounds. However, the presence of these tags downstream of the former weir site may also represent dead chum which washed downstream following spawning and may have actually spawned upstream of the weir site. Two of the 11 tags located downstream of the weir were located within 1 km of the weir and it is possible that these represent carcasses washing downstream. Unfortunately, without a stationary tower located at the weir site, there is no way of knowing if this was the case, so for the purposes of this report all estimates assume that fish found downstream of the weir spawned downstream of the weir. It is also important to note that the remaining nine tags were located 4.8 to 41 km downstream (river distance) of the weir and it is unlikely that tags would have drifted this far downstream.

A total of 4 tags were permanently removed in the local fishery and an additional 16 tags were not located during the tracking flights. The destinations of these 16 tags cannot be determined, and they should be removed from the data set when considering the proportion of chum salmon that spawned upstream of the former Fishing Branch River weir site.

If the total number of tagged chum salmon is adjusted to remove the tags that were removed in the local Old Crow fishery (4 tags), the tagged chum that were found near Old Crow (8 tags; assumed harvested or died as a result of tagging) and the tagged chum that were not found during tracking flights (16 tags), 65 tagged chum salmon remain. The proportion of this adjusted total number of tagged chum salmon that were located upstream of the former Fishing Branch River weir is 46.1% (Table 8).

Table 8. Summary of locations of chum salmon radio tagged in the Porcupine River in 2014; tags that were removed in the local fishery, found in the vicinity of Old Crow or not found during tracking are no included.

Radio Tag Location	Number of Tags Detected	Proportion of adjusted total (%)
South Fork of the Fishing Branch River, upstream of the former weir site	30	46.1
South Fork of the Fishing Branch River, downstream of the former weir site	11	16.9
Porcupine River mainstem, from Whitestone Confluence to Schaeffer Creek	8	12.3
Porcupine River mainstem, from Schaeffer Creek to mouth of Bell River	5	7.7
Porcupine River mainstem, between mouth of Driftwood River and David Lord Creek	1	1.5
Crow River	5	7.7
Bell River, including Rock River and upper Bell River	4	6.1
Bluefish River	1	1.5
TOTAL	**65**	**-**

4.3 ESTIMATE OF CHUM SALMON PASSAGE AT THE FISHING BRANCH WEIR

With knowledge of the total number of fish passing by the tagging site (via concurrent sonar operation) the percentages of tags found in a particular area can be used to roughly estimate the numbers at spawning destinations. . This apportionment provides a means to estimate the number of fish that spawned above the site of the former Fishing Branch River weir in 2014, which was a primary goal of this project.

A total of 16,892 chum salmon were estimated to have passed the Old Crow sonar site between August 12 and October 1, 2014 (Tanner pers. comm. 2015). However, 1,806 chum salmon were captured in the local VGG subsistence fishery, upstream of Old Crow (Tanner pers. comm. 2015). Although some of these chum salmon were undoubtedly not destined for spawning areas upstream of the former weir site, there is no reliable way to partition the local harvest by spawning destination. Removing the entire local harvest upstream of the sonar from the final sonar count is the best means to adjust the sonar count for harvest. If these chum salmon are removed from the sonar estimate, a harvest adjusted total of 15,086 chum salmon are estimated to have migrated to spawning areas upstream of the sonar site. An estimate of the proportion of this harvest adjusted sonar count that spawned upstream of the weir is obtained by multiplying 15,086 chum salmon by the adjusted percentage of radio tagged fish that were found upstream of the Fishing Branch weir (46.1%). This calculation indicates that an estimated 6,955 chum salmon spawned upstream of the site of the former Fishing Branch River weir during 2014.

4.4 COMPARISON TO HISTORICAL DATA

4.4.1 Tag Recovery from Mark-Recapture/CPUE Data

The proportion of radio tagged chum salmon that were relocated upstream of the Fishing Branch River in 2014 (46.1%) can be compared to the historical tag recovery rates from the chum salmon Mark Recapture/CPUE index programs (EDI 2011). During the Mark Recapture/CPUE index programs chum salmon tagging was conducted at a site 23 km downstream of Old Crow and there are no known chum salmon spawning areas between this site and Porcupine River Sonar site. As shown in Table 9, the tag recovery rate over 7 years of Mark-Recapture and CPUE index programs ranged from 35% to 88%, with an average of 56.4 % (EDI 2011). Although these programs used very different methods from the recent telemetry based studies, the 46.1% tag detection rate from the 2014 radio telemetry is within the lower end of the historical range of the Mark-Recapture and CPUE index programs.

Table 9. Relative proportion of tagged chum salmon counted at the Fishing Branch River weir (Data Source: EDI 2011).

Year	Tags Counted Through the Fishing Branch River Weir (%)[A]
2003	88
2004	59
2005	71
2006	- [B]
2007	56
2008	45
2009	35
2010	41
Average Tag Recovery Rate	**56.4**

[A] Percentages are calculated after the removal of tags by the Old Crow aboriginal fishery and are corrected for fish that may have crossed the weir while it was not operating.

[B] During 2006, the weir was not operated continuously due to flooding and as such a reliable proportion of tags could not be obtained.

4.4.2 2011 to 2014 Weir, Sonar and Telemetry Data Comparisons

The estimated proportion of chum salmon that were counted at the Porcupine River sonar site and spawned upstream of the Fishing Branch weir in 2014 can be compared to results of the 2013 telemetry program in addition to the concurrent sonar and weir counts from 2011 and 2012. In 2011 and 2012, the Porcupine River sonar operated concurrently with the Fishing Branch River weir. Table 10 compares counts from the sonar and weir sites for the periods in 2011 and 2012 when the two programs were operating simultaneously. The average travel times of chum salmon from the sonar to the weir site were 9 and 13 days in 2011 and 2012, respectively (EDI 2012, EDI 2013). These travel time offsets were used to compare counts from the two programs. For 2013 and 2014, tag relocations upstream of the weir were used in place of weir counts. Despite the difference in the methods of estimation, a similar percentage of chum salmon passed the Porcupine Sonar site in 2011, 2012 and 2013 and were subsequently counted/detected upstream of the site of the Fishing Branch River (Table 10). In 2014, there was a notable decrease in the proportion of tagged fish relocated above the former weir site. These results are consistent with ground observations at the Fishing Branch River spawning grounds which indicated that there were very low numbers of spawners during October 2014 (Timpany Pers. Comm. 2015).

Table 10. Comparison of harvest adjusted Porcupine River sonar and Fishing Branch River weir chum salmon passage estimates from 2011 and 2012 (EDI 2012, EDI 2013), 2013 radio telemetry data (EDI 2014b) and 2014 telemetry data. The Fishing Branch weir estimates for 2013 and 2014 are derived from the adjusted proportion of radio tag detections upstream of the weir site.

Program Year	Harvest Adjusted Porcupine River Sonar Estimate	Equivalent Fishing Branch River Weir Estimate	Percentage of Sonar Estimate
2011	10,938	8,574	78.3
2012	27,793	21,561	77.5
2013	34,235	25,368	74.1
2014	15,086	7,196	46.1
Averages	**22,013**	**15,675**	**69.0**

4.5 RECAPTURE OF TAGS IN LOCAL FISHERY

Of chum salmon radio tagged during the 2014 program only 6 individuals (6.3% of tagged fish) were captured in the local VGFN subsistence fishery near Old Crow, 2 of which were captured early enough in the program to redeploy the esophageal implant tags into other fish. The 2013 radio tagging program saw a high proportion (31%) of tags recaptured in the local VGFN subsistence fishery (Table 11). In 2014, the recapture rate was much lower (6.3%) perhaps due to the transition to internal esophageal implant tags and/or a reduced amount of fishing harvest by local subsistence fishers and differences in fishing locations used by local fishers. It is also important that the relative amount of harvest of chum salmon was higher in 2014 than 2013 and this may be due to a closure of the drainage wide Chinook subsistence fishery which resulted in local fishers harvesting more chum.

Table 11. Comparison of the recapture rate of radio tagged chum salmon to overall capture of chum in salmon the VGG subsistence fishery upstream of the Porcupine River sonar site in 2013 and 2014.

Program Year		Total Count	Number Recaptured in VGFN subsistence fishery	Proportion of Total Count (%)
2013	Chum Salmon Sonar Interpolated Total	35,615	1,380	3.9
	Radio Tagged Chum Salmon	94	29	31
2014	Chum Salmon Sonar Interpolated Total	16,892	1,806	10.7
	Radio Tagged Chum Salmon	95[A]	6[A]	6

[A] The two tags that were captured and redeployed have been added to the totals to make the proportions harvested by fishery more comparable to 2013 when none of the tags were reapplied.

As in 2013, the tagging location of the Porcupine River Sonar site was downstream of a number of local VGG subsistence fishing sites. This site was chosen for several important reasons. First, the primary goal of this project was to estimate the number of chum salmon that spawned upstream of the Fishing Branch River weir site in 2013. This estimate was derived from chum sonar counts and tag relocation rates upstream of the former weir site. To produce this estimate, the tagging had to be conducted at or near the sonar site. Second, tagging could not be conducted upstream of the Old Crow River, as it is suspected to support chum salmon spawning (Anderton and Frost 2003); this suspicion was confirmed by the 2014 tag relocations. If tagging were conducted upstream of the mouth of the Crow River, there would be no way to determine the proportion of chum that were counted at the sonar site and spawned in the Old Crow River.

4.6 FATES OF MISSING RADIO TAGS

Fourteen radio tagged chum salmon were not detected during the 2014 tracking flights. There are several possible theories that may explain this occurrence. Tag failure could account for a few of the missing tags. Hardware failures on VHF radio tags of the type used in the 2014 program are rare; according to manufacturer data, the failure rate is approximately 2% (Adsem pers. comm. 2013). Although it is understood to be a very infrequent occurrence, a 2% tag failure rate could account for some of the missing tags that were applied in 2014.

Another possibility is that a small number of tagged chum salmon may have migrated upstream into un-surveyed areas of the Porcupine River watershed. This seems unlikely to account for a large number of the missing tags, particularly given the very extensive aerial telemetry flights conducted during 2014. It is very unlikely that tags were missed within the portion of the watershed surveyed. The telemetry equipment used often detected tags a considerable distance (5-10 km) away from the actual tag location. Areas with a high density of tags (Fishing Branch River) were also flown a number of times (once on each frequency) to ensure that all tags were detected. Damage to tags could also have occurred from predators consuming tagged salmon on the spawning grounds.

A final and most likely explanation for the missing tags is downstream migration following tagging. For example, during 2013, a radio tagged chum was recaptured near the mouth of the Campbell River in Alaska, approximately 130 km downstream of the tagging site. Furthermore, radio tags were relocated in the Bluefish River during 2013 (1 tag) and 2014 (1 tag) despite being 40 km downstream of the tagging site. The same pattern was observed during the 2005 and 2007 coho salmon telemetry projects when small numbers of tags were relocated in the Bluefish River despite being 35 km downstream of the tagging site (Anderton and Schonewille 2006, EDI 2008). Collectively, this information suggests that it is not uncommon for tagged fish to migrate downstream after capture, the reason for which may be due to natural migration patterns (overshooting) or possible handling induced stress.

5 CONCLUSIONS

The 2014 Porcupine River chum salmon telemetry program provided valuable insights into the relationship between sonar counts and the number of chum salmon that spawn upstream of the former Fishing Branch River weir site. This project confirmed that the areas of the Fishing Branch River upstream of the former weir site remain the primary spawning destination for chum salmon in the Porcupine River watershed. This program also provided the fourth year of comparative escapement estimates allowing for the calculation of the average proportion of chum salmon that migrate past the Porcupine River sonar site en-route to spawning grounds upstream of the former weir site. In addition, the project provided new information on chum salmon spawning in other watersheds within the Porcupine River which were previously un-documented.

Esophageal implant tags likely helped reduce the number of tags captured in the VGFN subsistence fishery. However, as run size and fishing effort differed between 2013 and 2014, it is impossible to solely attribute this success to the change in tag type alone. In 2014 there was a much higher rate of tags that were not relocated during aerial telemetry surveys. The reasons tags are not located remain uncertain and future projects should take the likelihood of significant tag loss into consideration during the planning stages.

The incorporation of two stationary telemetry towers during the 2015 chum telemetry program would assist in determining the fate of radio tags. A tower placed downstream of the tagging site would provide quantitative data on tag drop outs and/or downstream migration after tagging. A second tower located at the site of the former Fishing Branch weir site could provide a more accurate passage rate of tagged chum and would determine if the tags found downstream of the weir site are a result of carcasses washing downstream after spawning.

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6.1 Personal Communications

Adsem, J. Advanced Telemetry Systems Territory Manager & Project Consultant. Phone call to the author. January 7, 2014.

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Timpany, P. Operator of Bear Cave Mountain Eco-Adventures. Discussions with EDI, January 2015.

APPENDIX A. TAG FREQUENCY LIST PROVIDED BY ADVANCED TELEMTRY SYSTEMS

ADVANCED TELEMETRY SYSTEMS
470 FIRST AVE N • ISANTI, MN 55040-7123
763-444-9267 • 763-444-9384 fax
email:sales@atstrack.com • www.atstrack.com

Transmitter Data Sheet

Job #: 81902
Ship Date: 5/28/2014

Customer Info

Edynamics
Attn: Ben Snow
2195 Second Ave.

Whitehorse YT Y1A 3T8
CANADA

Serial #	Quantity	Model#	Description	Species
01	75	F1840B	Implant Transmitter	Chum
		ProgDesc:	Mortality: 24 hr	

Pulse Rate: 45.8 ppm **Pulse Width:** 30 ms **Weight:** 22 gr **Warranty Life:** 63 days **Battery Cap Life:** 127 days

164.187	164.187	164.187	164.187
A-0/830/45	A-1/850/55	A-11/1060/155	A-12/1000/125
164.187	164.187	164.187	164.187
A-13/1090/175	A-14/1130/185	A-15/1140/195	A-16/1160/205
164.187	164.187	164.187	164.187
A-17/1180/215	A-18/1190/225	A-19/1220/235	A-2/890/65
164.187	164.187	164.187	164.187
A-20/1240/245	A-21/1270/255	A-22/1290/265	A-23/1300/275
164.187	164.187	164.187	164.187
A-24/1310/285	A-26/990/115	A-3/910/75	A-4/920/85
164.187	164.187	164.187	164.187
A-5/940/95	A-6/970/105	A-75/1030/145	A-8/1070/165
164.187	164.206	164.206	164.206
A-9/1010/135	A-0/830/45	A-1/850/55	A-11/1060/155
164.206	164.206	164.206	164.206
A-12/1000/125	A-13/1090/175	A-14/1130/185	A-15/1140/195
164.206	164.206	164.206	164.206
A-16/1160/205	A-17/1180/215	A-18/1190/225	A-19/1220/235
164.206	164.206	164.206	164.206
A-2/890/65	A-20/1240/245	A-21/1270/255	A-22/1290/265
164.206	164.206	164.206	164.206
A-23/1300/275	A-24/1310/285	A-26/990/115	A-3/910/75
164.206	164.206	164.206	164.206
A-4/920/85	A-5/940/95	A-6/970/105	A-75/1030/145
164.206	164.206	164.236	164.236
A-8/1070/165	A-9/1010/135	A-0/830/45	A-1/850/55
164.236	164.236	164.236	164.236
A-11/1060/155	A-12/1000/125	A-13/1090/175	A-14/1130/185
164.236	164.236	164.236	164.236
A-15/1140/195	A-16/1160/205	A-17/1180/215	A-18/1190/225
164.236	164.236	164.236	164.236
A-19/1220/235	A-2/890/65	A-20/1240/245	A-21/1270/255
164.236	164.236	164.236	164.236
A-22/1290/265	A-23/1300/275	A-24/1310/285	A-26/990/115
164.236	164.236	164.236	164.236
A-3/910/75	A-4/920/85	A-5/940/95	A-6/970/105
164.236	164.236	164.236	
A-75/1030/145	A-8/1070/165	A-9/1010/135	

Serial #	Quantity	Model#	Description	Species
02	20	F1840B	Implant Transmitter	Chum
		ProgDesc:	Mortality: 24 hr	

164.506	164.506	164.506	164.506
A-11/1060/155	A-12/1000/125	A-13/1090/175	A-14/1130/185
164.506	164.506	164.506	164.506
A-15/1140/195	A-16/1160/205	A-17/1180/215	A-18/1190/225
164.506	164.506	164.506	164.506
A-19/1220/235	A-20/1240/245	A-21/1270/255	A-22/1290/265
164.506	164.506	164.506	164.506
A-23/1300/275	A-24/1310/285	A-3/910/75	A-4/920/85
164.506	164.506	164.506	164.506
A-5/940/95	A-6/970/105	A-8/1070/165	A-9/1010/135

APPENDIX B. DRIFT AND SET NETTING EFFORT & FISH CAPTURE DATA

Table B1. Summary of chum salmon radio tagged during 2014. Note that sampling data (sex, fork length) from August 25 to September 7 was partially lost in the field.

Tag Number	Tag Frequency	Tag Code	Date Applied	Location	Reapplication	Floy Tag #	Sex	Fork Length (cm)	Comments
1	164.187	0	25-Aug-14	sonar - LB	No		M		
2	164.187	1	25-Aug-14	sonar - LB	No		M		
3	164.187	2	25-Aug-14	sonar - LB	No		M		
4	164.187	3	25-Aug-14	sonar - LB	No		M		
5	164.187	4	26-Aug-14	sonar - LB	No		M		
6	164.187	5	27-Aug-14	sonar - LB	No				
7	164.187	6	27-Aug-14	sonar - LB	No				
8	164.187	8	27-Aug-14	sonar - LB	No				
9	164.187	9	27-Aug-14	sonar - LB	No				
10	164.187	11	27-Aug-14	sonar - LB	No				
11	164.187	12	27-Aug-14	sonar - LB	No				
12	164.187	13	27-Aug-14	sonar - LB	No				
13	164.187	14	28-Aug-14	sonar - LB	No				
14	164.187	15	28-Aug-14	sonar - LB	No				
15	164.187	16	29-Aug-14	sonar - LB	No				
16	164.187	17	30-Aug-14	sonar - LB	No				
17	164.187	18	30-Aug-14	sonar - LB	No				
18	164.187	19	30-Aug-14	sonar - LB	No				
19	164.187	20	31-Aug-14	sonar - LB	No				
20	164.187	21	2-Sep-14	sonar - LB	No				
21	164.187	22	7-Sep-14	sonar - LB	No				
22	164.187	23	7-Sep-14	sonar - LB	No				
23	164.187	24	7-Sep-14	sonar - LB	No				
24	164.187	26	9-Sep-14	sonar - LB	No	B04542	M	59	recaptured by local fisher
24	164.187	26	18-Sep-14	sonar - LB	Yes	B04772	M	74.5	redeployment of recaptured tag
25	164.187	75	9-Sep-14	sonar - LB	No	B04499	M	65	
26	164.206	0	10-Sep-14	sonar - LB	No	B04540	M	76	
27	164.206	1	10-Sep-14	sonar - LB	No	B04507	M	66	
28	164.206	2	10-Sep-14	sonar - LB	No	B04508	M	64	
29	164.206	3	10-Sep-14	sonar - LB	No	B04545	M	65	
30	164.206	4	10-Sep-14	sonar - RB	No	B04509	M	64	
31	164.206	5	11-Sep-14	sonar - LB	No	B04639	M	61	
32	164.206	6	11-Sep-14	sonar - LB	No	B04640	M	65	
33	164.206	8							tag lost in the field
34	164.206	9	11-Sep-14	sonar - LB	No	B04641	F	62	
35	164.206	11	11-Sep-14	sonar - RB	No	B04590	M	60	
36	164.206	12	15-Sep-14	sonar - RB	No	B04594	F	62	
37	164.206	13	15-Sep-14	sonar - LB	No	B04638	M	65	recaptured by local fisher
37	164.206	13	18-Sep-14	sonar - LB	Yes	B04510	F	63	redeployment of recaptured tag

Tag Number	Tag Frequency	Tag Code	Date Applied	Location	Reapplication	Floy Tag #	Sex	Fork Length (cm)	Comments
38	164.206	14	16-Sep-14	sonar - LB	No	B04595	F	59	
39	164.206	15	16-Sep-14	sonar - LB	No	B04636	F	62	
40	164.206	16	16-Sep-14	sonar - LB	No	B04633	F	60	
41	164.206	17	16-Sep-14	sonar - LB	No	B04631	F	59	
42	164.206	18	17-Sep-14	sonar - LB	No	B04635	F	66	
43	164.206	19	17-Sep-14	sonar - LB	No	B04546	M	72	
44	164.206	20	17-Sep-14	sonar - LB	No	B04632	F	65	
45	164.206	21	17-Sep-14	sonar - LB	No	B04637	F	67	
46	164.206	22	17-Sep-14	sonar - LB	No	B04634	F	62	
47	164.206	23	18-Sep-14	sonar - LB	No	B04642	M	76.5	
48	164.206	24	19-Sep-14	sonar - LB	No	B04547	F	72	
49	164.206	26	19-Sep-14	sonar - LB	No	B04630	F	62	
50	164.206	75	19-Sep-14	sonar - LB	No	B04493	M	75.5	
51	164.236	0	21-Sep-14	sonar - LB	No	B04643	M	76	
52	164.236	1	21-Sep-14	sonar - LB	No	B04673	F	63	
53	164.236	2	22-Sep-14	sonar - LB	No	BO4629	M	71	
54	164.236	3	22-Sep-14	sonar - LB	No	BO4722	M	63	
55	164.236	4	22-Sep-14	sonar - LB	No	BO4721	F	59	
56	164.236	5	22-Sep-14	sonar - LB	No	BO4628	M	65	
57	164.236	6	22-Sep-14	sonar - LB	No	BO4644	F	58	
58	164.236	8	23-Sep-14	sonar - LB	No	BO4580	F	65	
59	164.236	9	23-Sep-14	sonar - LB	No	BO4490	M	70	
60	164.236	11	23-Sep-14	sonar - LB	No	BO4481	M	68	
61	164.236	12	24-Sep-14	sonar - LB	No	BO4579	M	67	
62	164.236	13	24-Sep-14	sonar - LB	No	BO4582	F	64	
63	164.236	14	24-Sep-14	sonar - LB	No	BO4582	M	62	Duplicate spaghetti tag
64	164.236	15	25-Sep-14	sonar - LB	No	BO4539	F	60	
65	164.236	16	25-Sep-14	sonar - LB	No	BO4483	F	61	
66	164.236	17	25-Sep-14	sonar - LB	No	B04513	M	68	
67	164.236	18	26-Sep-14	sonar - LB	No	BO4771	M	72	
68	164.236	19	26-Sep-14	sonar - LB	No	BO4770	F	70	
69	164.236	20	26-Sep-14	sonar - LB	No	BO4578	F	64	
70	164.236	21	27-Sep-14	sonar - LB	No	BO4538	M	67	
71	164.236	22	27-Sep-14	sonar - LB	No	B04482	M	67	
72	164.236	23	27-Sep-14	sonar - LB	No	B04597	F	60	
73	164.236	24	29-Sep-14	sonar - LB	No	BO4537	F	63	
74	164.236	26	29-Sep-14	sonar - LB	No	B04514	F	61	
75	164.236	75							test tag, not applied
76	164.506	3	19-Sep-14	sonar - LB	No	B04492	M	70	
77	164.506	4	20-Sep-14	sonar - LB	No	B04549	M	76	
78	164.506	5	20-Sep-14	sonar - LB	No	B04724	F	66	
79	164.506	6	20-Sep-14	sonar - LB	No	B04548	F	59	

Tag Number	Tag Frequency	Tag Code	Date Applied	Location	Reapplication	Floy Tag #	Sex	Fork Length (cm)	Comments
80	164.506	8	20-Sep-14	sonar - LB	No	B04546	F	63	
81	164.506	9	20-Sep-14	sonar - LB	No	B04725	M	71	
82	164.506	11	20-Sep-14	sonar - LB	No	B04727	M	70	
83	164.506	12	20-Sep-14	sonar - LB	No	B04726	M	66	
84	164.506	13	21-Sep-14	sonar - LB	No	B04723	M	66	
85	164.506	14	21-Sep-14	sonar - LB	No	B04728	F	60	
86	164.506	15	21-Sep-14	sonar - LB	No	B04596	F	62	
87	164.506	16	21-Sep-14	sonar - LB	No	B04479	F	62	
88	164.506	17	21-Sep-14	sonar - LB	No	B04550	F	61	
89	164.506	18	21-Sep-14	sonar - LB	No	B04774	F	64	
90	164.506	19	21-Sep-14	sonar - LB	No	B04511	M		
91	164.506	20	21-Sep-14	sonar - LB	No	B04729	F	62	
92	164.506	21	21-Sep-14	sonar - LB	No	B04773	F	66	
93	164.506	22	21-Sep-14	sonar - LB	No	B04512	M	67	
94	164.506	23	21-Sep-14	sonar - LB	No	B04551	M	70	
95	164.506	24	21-Sep-14	sonar - LB	No	B04730	F	57	

APPENDIX C. LOCATION OF TAGS DETECTED DURING AERIAL TELEMETRY

Table C1. Summary of radio tagged chum salmon relocated during aerial telemetry flights (October 23 – 26, 2014).

Frequency	Code	Location	Mortality Status	UTM Coordinates (Zone 07W)	
				Easting	Northing
164.187	1	South Fork of the Fishing Branch River, downstream of the former weir site	Dead	596199	7368414
164.187	2	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	574913	7379850
164.187	4	South Fork of the Fishing Branch River, upstream of the former weir site	Alive	572654	7377214
164.187	5	Porcupine River mainstem, from Whitestone Confluence to Schaeffer Creek	Alive	618694	7384240
164.187	6	Porcupine River mainstem, from Whitestone Confluence to Schaeffer Creek	Dead	614151	7378715
164.187	8	South Fork of the Fishing Branch River, downstream of the former weir site	Dead	587711	7372285
164.187	9	Porcupine River mainstem, from Schaeffer Creek to mouth of Bell River	Dead	389369	7431596
164.187	11	Old Crow, and near vicinity	Alive	550006	7496332
164.187	12	lower Bell River	Alive	402496	7475188
164.187	14	Porcupine River mainstem, between mouth of Driftwood River and David Lord Creek	Alive	592326	7499472
164.187	15	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	574225	7379802
164.187	16	South Fork of the Fishing Branch River, upstream of the former weir site	Alive	574848	7379843
164.187	17	Old Crow, and near vicinity	Dead	549224	7495755
164.187	18	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	574003	7379770
164.187	19	Rock River (Bell River watershed)	Dead	424403	7433796
164.187	21	Porcupine River mainstem, from Whitestone Confluence to Schaeffer Creek	Alive	619066	7385241
164.187	22	Porcupine River mainstem, from Whitestone Confluence to Schaeffer Creek	Dead	619085	7385277
164.187	23	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	575592	7379788
164.187	24	Porcupine River mainstem, from Whitestone Confluence to Schaeffer Creek	Alive	619008	7385126
164.187	26	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	575182	7379861
164.187	75	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	572662	7377186
164.206	0	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	575586	7379897
164.206	1	South Fork of the Fishing Branch River, upstream of the former weir site	Unclear	572261	7377856
164.206	2	South Fork of the Fishing Branch River, downstream of the former weir site	Dead	593036	7369749
164.206	4	Bluefish River	Dead	528138	7474150
164.206	5	South Fork of the Fishing Branch River, upstream of the former weir site	Unclear	576292	7379811
164.206	9	upper Old Crow River, near YT/AK border	Dead	501117	7566152
164.206	11	South Fork of the Fishing Branch River, downstream of the former weir site	Dead	584935	7375718
164.206	12	South Fork of the Fishing Branch River, upstream of the former weir site	Alive	574336	7379930
164.206	13	South Fork of the Fishing Branch River, downstream of the former weir site	Dead	580371	7378504
164.206	14	Porcupine River mainstem, from Whitestone Confluence to Schaeffer Creek	Alive	615168	7379154
164.206	15	South Fork of the Fishing Branch River, upstream of the former weir site	Unclear	574234	7379921
164.206	16	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	572528	7377369
164.206	18	South Fork of the Fishing Branch River, upstream of the former weir site	Alive	576786	7379822
164.206	19	South Fork of the Fishing Branch River, downstream of the former weir site	Dead	578412	7379567
164.206	20	South Fork of the Fishing Branch River, downstream of the former weir site	Dead	589294	7370690
164.206	21	South Fork of the Fishing Branch River, downstream of the former weir site	Dead	599473	7368249
164.206	22	Porcupine River mainstem, from Schaeffer Creek to mouth of Bell River	Alive	386173	7443598

164.206	23	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	575691	7379885
164.206	24	South Fork of the Fishing Branch River, upstream of the former weir site	Alive	575796	7379872
164.206	26	Porcupine River mainstem, from Whitestone Confluence to Schaeffer Creek	Dead	619379	7385748
164.206	75	South Fork of the Fishing Branch River, downstream of the former weir site	Alive	578412	7379567
164.236	0	lower Old Crow River, near Schaeffer Mountain	Alive	544966	7519794
164.236	1	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	575280	7379728
164.236	3	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	572543	7377361
164.236	5	Old Crow, and near vicinity	Alive	549896	7495675
164.236	6	South Fork of the Fishing Branch River, downstream of the former weir site	Dead	596858	7368331
164.236	8	Porcupine River mainstem, from Whitestone Confluence to Schaeffer Creek	Alive	374618	7416952
164.236	9	upper Old Crow River, near YT/AK border	Alive	505412	7562705
164.236	11	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	575793	7379850
164.236	12	South Fork of the Fishing Branch River, downstream of the former weir site	Dead	586654	7373741
164.236	13	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	572623	7377238
164.236	14	Porcupine River mainstem, from Schaeffer Creek to mouth of Bell River	Dead	387997	7447501
164.236	17	Old Crow, and near vicinity	Dead	550620	7495496
164.236	18	Old Crow, and near vicinity	Alive	550564	7495417
164.236	19	Old Crow, and near vicinity	Alive	548792	7495702
164.236	20	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	577674	7379871
164.236	21	Old Crow, and near vicinity	Dead	549234	7495978
164.236	22	upper Bell River (above the Little Bell River)	Alive	426338	7512176
164.236	24	lower Old Crow River, near Schaeffer Mountain	Alive	544073	7517883
164.236	26	Old Crow, and near vicinity	Dead	549862	7495657
164.506	3	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	575181	7379880
164.506	4	South Fork of the Fishing Branch River, upstream of the former weir site	Alive	576901	7379758
164.506	6	upper Bell River (above the Little Bell River)	Alive	425646	7512476
164.506	11	South Fork of the Fishing Branch River, upstream of the former weir site	Dead	572572	7377139
164.506	12	South Fork of the Fishing Branch River, upstream of the former weir site	Alive	575046	7379894
164.506	13	South Fork of the Fishing Branch River, upstream of the former weir site	Alive	572385	7378267
164.506	14	Porcupine River mainstem, from Whitestone Confluence to Schaeffer Creek	Alive	370309	7403742
164.506	15	South Fork of the Fishing Branch River, upstream of the former weir site	Alive	574845	7379908
164.506	16	South Fork of the Fishing Branch River, upstream of the former weir site	Alive	575485	7379829
164.506	17	South Fork of the Fishing Branch River, upstream of the former weir site	Alive	573874	7379777
164.506	20	Old Crow River near Black Fox Creek	Dead	555943	7549429
164.506	24	Porcupine River mainstem, from Whitestone Confluence to Schaeffer Creek	Alive	613281	7378519